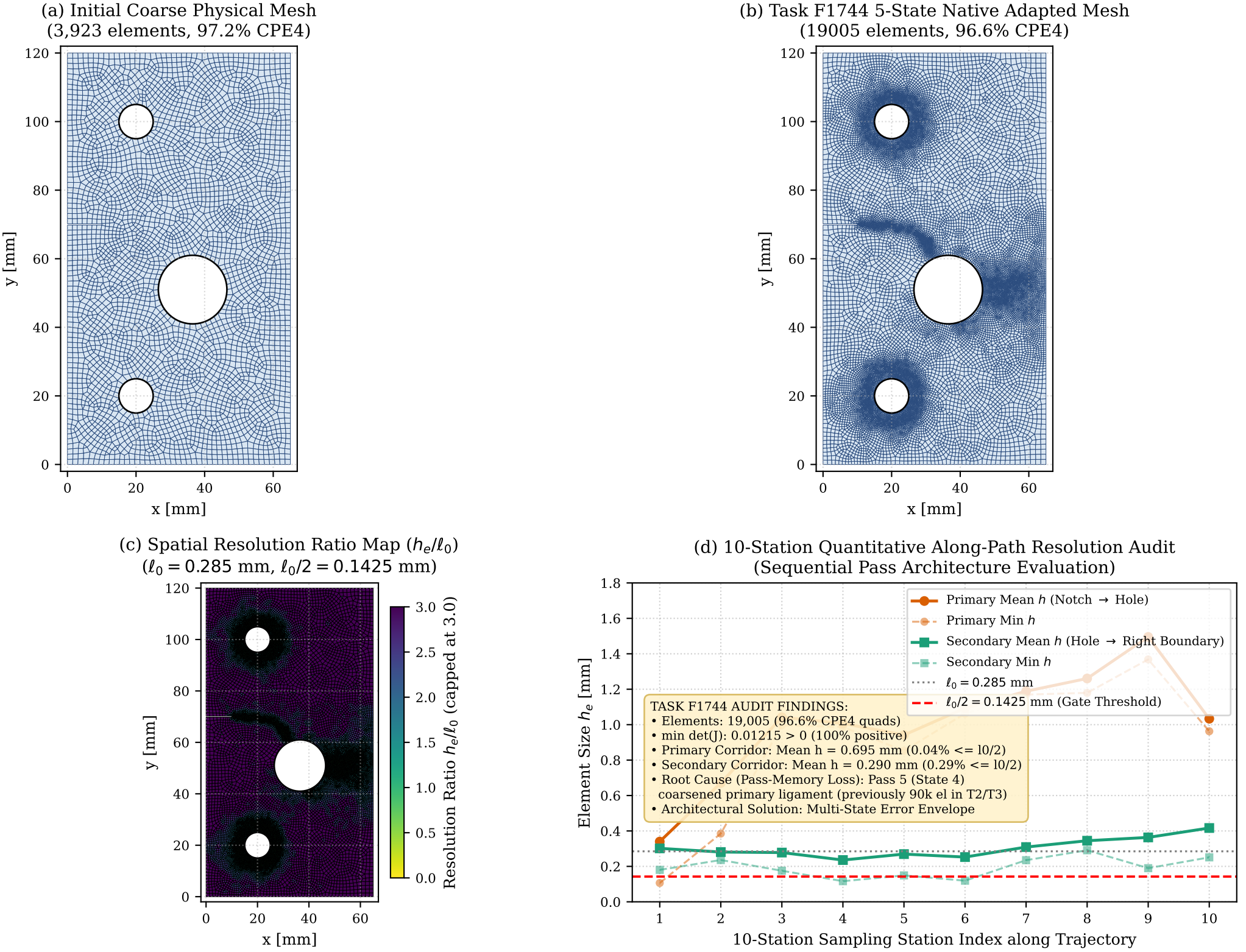

(a) Initial Coarse Physical Mesh
(3,923 elements, 97.2% CPE4)
y [mm]
x [mm]
(b) Task F1744 5-State Native Adapted Mesh
(19005 elements, 96.6% CPE4)
y [mm]
x [mm]
(c) Spatial Resolution Ratio Map (h_e/ℓ_0)
($\ell_0 = 0.285$ mm, $\ell_0/2 = 0.1425$ mm)
y [mm]
x [mm]
Resolution Ratio h_e/ℓ_0 (capped at 3.0)
(d) 10-Station Quantitative Along-Path Resolution Audit
(Sequential Pass Architecture Evaluation)
Primary Mean h (Notch → Hole)
Primary Min h
Secondary Mean h (Hole → Right Boundary)
Secondary Min h
$\ell_0 = 0.285$ mm
$\ell_0/2 = 0.1425$ mm (Gate Threshold)
TASK F1744 AUDIT FINDINGS:
• Elements: 19,005 (96.6% CPE4 quads)
• min det(J): 0.01215 > 0 (100% positive)
• Primary Corridor: Mean h = 0.695 mm (0.04% <= l0/2)
• Secondary Corridor: Mean h = 0.290 mm (0.29% <= l0/2)
• Root Cause (Pass-Memory Loss): Pass 5 (State 4)
coarsened primary ligament (previously 90k el in T2/T3)
• Architectural Solution: Multi-State Error Envelope
Element Size h_e [mm]
10-Station Sampling Station Index along Trajectory